

Name: _____



2014

Mathematics Extension 2 Trial HSC Exam

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen.
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I

pages 2 – 5

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II

pages 6 – 16

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

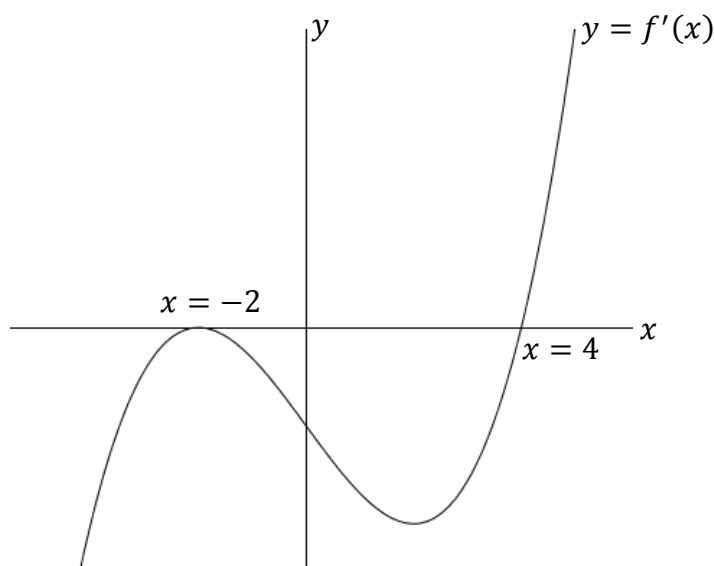
Use the multiple-choice answer sheet for Questions 1-10.

- 1 Which of the following is equivalent to

$$\int x \sec^2(x^2) dx$$

- (A) $2 \tan(x^2) + C$
- (B) $\frac{1}{2} \tan(x^2) + C$
- (C) $\frac{1}{3} \tan(x^2) + C$
- (D) $3 \tan(x^2) + C$

- 2 From the graph of $y = f'(x)$ drawn, which could be the equation of $y = f(x)$



- (A) $(x - 6)(x - 2)^3$
- (B) $(x + 6)(x - 2)^3$
- (C) $(x - 6)(x + 2)^3$
- (D) $(x + 6)(x + 2)^3$

3 The $\sqrt{-3 + 4i}$ is

- (A) $2 + i$
- (B) $1 + 2i$
- (C) $2 - i$
- (D) $1 - 2i$

4 A conic section has foci $S = (3,0)$ and $S' = (-3,0)$ and vertices $(2,0)$ and $(-2,0)$.

The equation of the conic is

- (A) $\frac{x^2}{5} + \frac{y}{4} = 1$
- (B) $\frac{x^2}{4} + \frac{y^2}{5} = 1$
- (C) $\frac{x^2}{4} - \frac{y^2}{5} = 1$
- (D) $\frac{x^2}{5} - \frac{y^2}{4} = 1$

5 For the function $g(x) = \tan^{-1}(e^x)$ the range is

- (A) $0 \leq y \leq \frac{\pi}{2}$
- (B) $0 \leq y < \frac{\pi}{2}$
- (C) $0 < y \leq \frac{\pi}{2}$
- (D) $0 < y < \frac{\pi}{2}$

6 If $e^x + e^y = 1$, which of the following is an expression for $\frac{dy}{dx}$?

- (A) $-e^{x-y}$
- (B) e^{x-y}
- (C) e^{y-x}
- (D) $-e^{y-x}$

- 7 The polynomial $P(x) = x^4 + ax^2 + bx + 28$ has a double root at $x = 2$.

What are the values of a and b ?

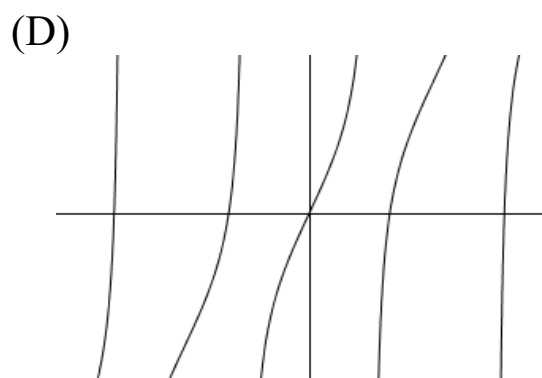
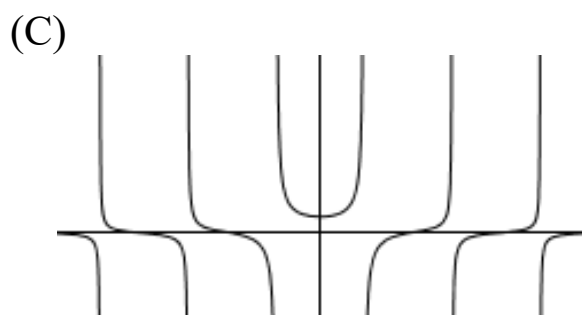
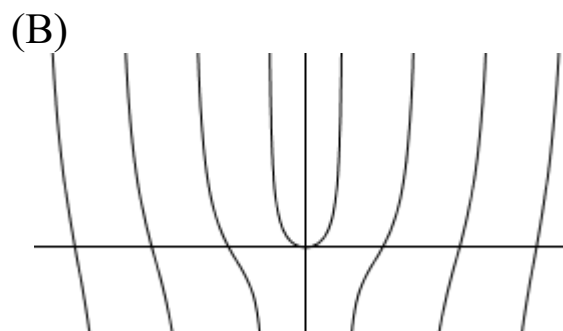
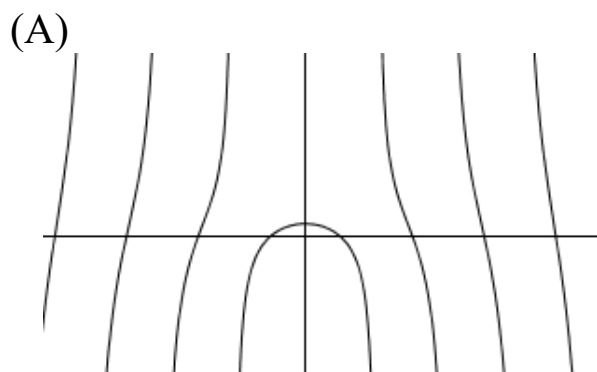
- (A) $a = -11$ and $b = -12$
- (B) $a = -5$ and $b = -12$
- (C) $a = -11$ and $b = 12$
- (D) $a = -5$ and $b = 12$

- 8 If ω is a non-real sixth root of -1 and ϕ is a non-real fifth root of 1 consider the following two statements:

- (I) $1 - \omega + \omega^2 - \omega^3 + \omega^4 - \omega^5 = 0$
- (II) $1 + \phi + \phi^2 + \phi^3 + \phi^4 = 0$

- (A) Both (I) and (II) are correct
- (B) Only (I) is correct
- (C) Only (II) is correct
- (D) Neither is correct

- 9 Which of the following best represents the graph of $g(x) = x \tan x$



- 10 Without evaluating the integrals, which one of the following integrals is greater than zero?

(A) $\int_{-1}^1 -e^x dx$

(B) $\int_{-1}^1 \frac{\sin^{-1} x}{x^2+1} dx$

(C) $\int_{-1}^1 \frac{\tan^{-1} x}{\cos x} dx$

(D) $\int_{-1}^1 e^{-x^2} dx$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) For the complex numbers $z = 1 + i$ and $w = 2 - 3i$ find:

- | | | |
|-------|------------------------------------|---|
| (i) | $\bar{z} - w$ | 1 |
| (ii) | zw | 1 |
| (iii) | Write z in modulus-argument form | 2 |

(b) Show that $z = \sqrt{2} \operatorname{cis} \left(\frac{\pi}{3} \right)$ is a solution of the equation 2
$$z^8 - 8z^2 = 0$$

(c) Find all real roots of the polynomial 3
$$P(x) = x^4 - x^3 - 4x^2 - 2x - 12,$$

given that one root is $\sqrt{2} i$.

Question 11 continues on page 7

- (d) Sketch the locus of z where the following conditions hold simultaneously **3**
 $0 \leq \arg(z - i) \leq \frac{2\pi}{3}$ and $|z - i| \leq 2$
- (e) In an Argand diagram the points P, Q and R represent complex numbers z_1, z_2 and $z_2 + i(z_2 - z_1)$ respectively.
- (i) Show that PQR is a right-angled isosceles triangle **2**
- (ii) Find in terms of z_1 and z_2 the complex number represented by the point S such that $PQRS$ is a square. **1**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) By completing the square find

2

$$\int \frac{1}{\sqrt{2 - (x^2 + 4x)}} dx$$

(b) Find

2

$$\int \frac{\sin^3 x}{\cos^2 x} dx$$

(c) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to evaluate

4

$$\int_0^{\frac{\pi}{2}} \frac{dx}{\cos x + \sin x + 1}$$

(d) (i) Find the values of A , B , C and D such that

2

$$\frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 + 1}$$

(ii) Hence find

2

$$\int \frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} dx$$

(e) Evaluate

3

$$\int_1^e \frac{\ln x}{\sqrt{x}} dx$$

Question 13 (15 marks) Use a SEPARATE writing booklet.

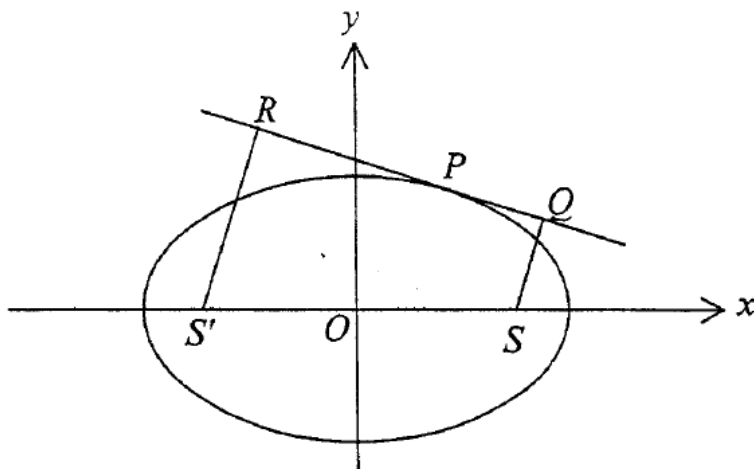
- (a) (i) Use the substitution $x = t - y$ where t is a constant to show 1

$$\int_0^t f(x)dx = \int_0^t f(t-x)dx$$

- (ii) Hence, evaluate 2

$$\int_0^1 x(1-x)^{2014}dx$$

(b)



- (i) Prove that the equation of the tangent to the ellipse 3

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

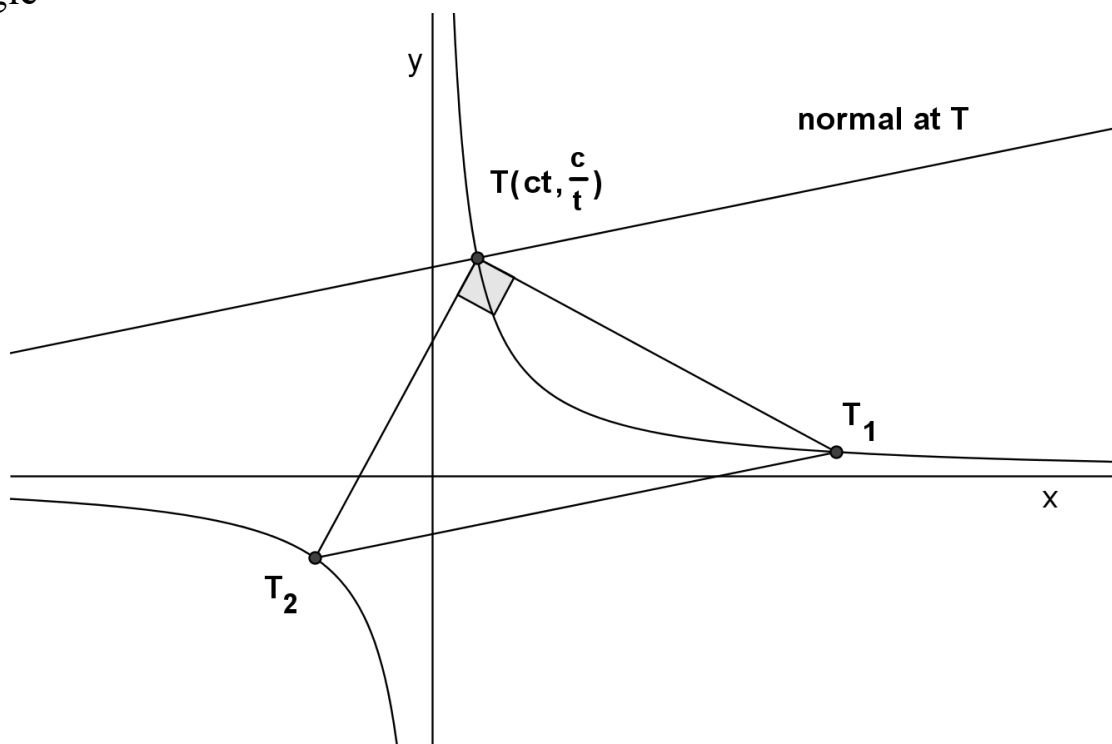
at the point $P(a \cos \theta, b \sin \theta)$ is

$$(b \cos \theta)x + (a \sin \theta)y - ab = 0$$

- (ii) Q and R are the feet of the perpendiculars to the tangent from the foci S and S' respectively 3
Prove that $SQ \times S'R = b^2$

Question 13 continues on page 10

- (c) As shown in the diagram below T_1 and T_2 are two points on the rectangular hyperbola $xy = c^2$ with parameters t_1 and t_2 respectively and T is a third point on it with parameter t such that $\angle T_1 T T_2$ is a right angle



- | | | |
|-------|---|----------|
| (i) | Show that gradient of $T_1 T$ is $-\frac{1}{t_1 t}$ and deduce that since $\angle T_1 T T_2$ is a right angle then $t^2 = -\frac{1}{t_1 t_2}$ | 3 |
| (ii) | Write down the gradient of $T_1 T_2$ | 1 |
| (iii) | Hence, prove that $T_1 T_2$ is parallel to the normal at T | 2 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

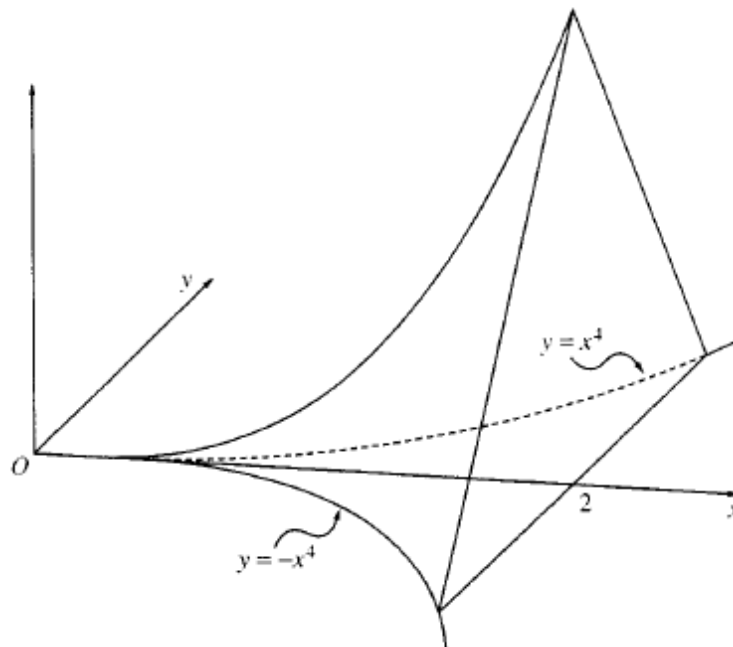
- (a) α, β and γ are the roots of the equation 2

$$x^3 - 6x^2 + 12x - 35 = 0$$

Form a cubic equation whose roots are $\alpha - 2$, $\beta - 2$ and $\gamma - 2$

- (b) By taking slices perpendicular to the axis of rotation find the volume of the solid generated by rotating the region bounded by the curve $y = (x - 2)^2$ and the line $y = x$ about the x -axis 3

- (c) The base of a solid is the region in the xy plane enclosed by the curves $y = x^4$, $y = -x^4$ and the line $x = 2$. Each cross-section perpendicular to the x -axis is an equilateral triangle

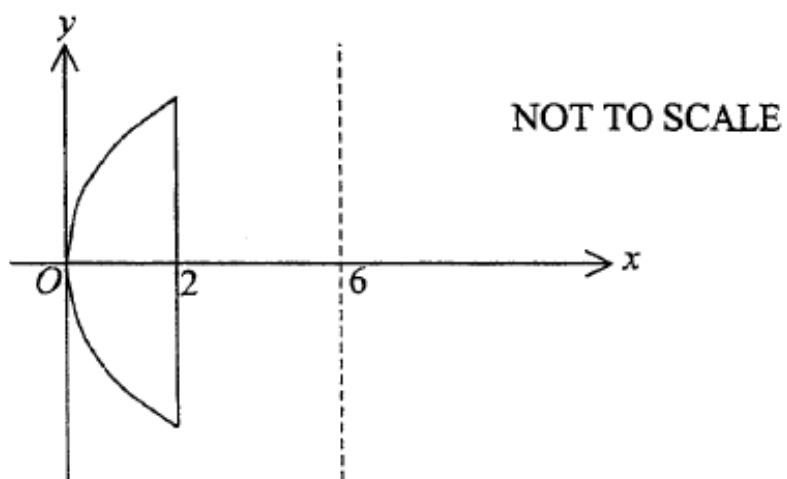


- (i) Show that the area of the triangular cross-section at $x = h$ is $\sqrt{3} h^8$ 2
- (ii) Hence, find the volume of the solid 3

Question 14 continues on page 12

(d)

5



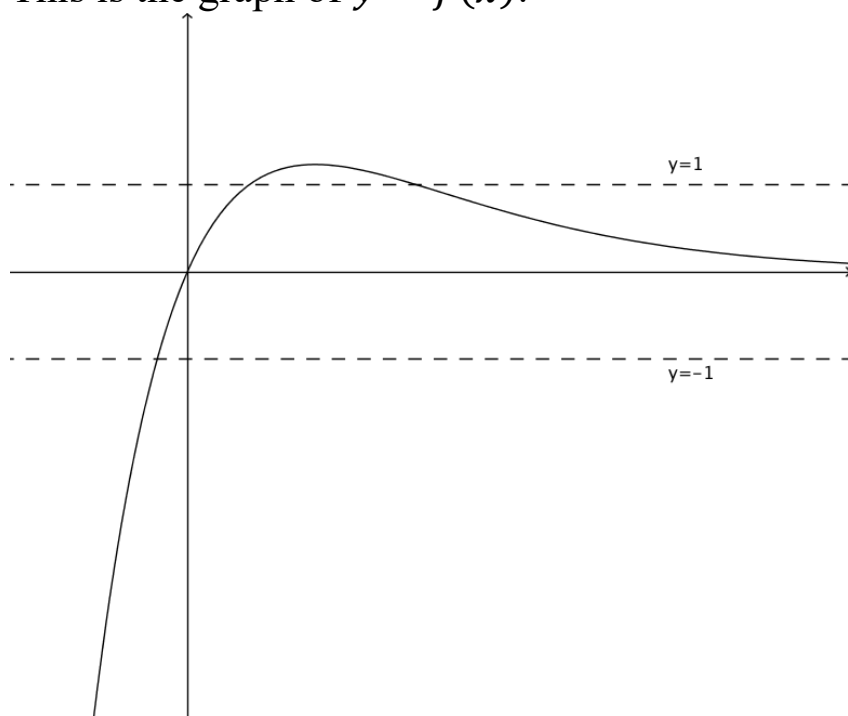
The region bounded by the parabola $y^2 = 4x$ and the line $x = 2$ is rotated about the line $x = 6$.

Using the method of cylindrical shells, find the volume of the solid formed.

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) This is the graph of $y = f(x)$:



On the separate graph answer sheet, sketch:

- | | | |
|-------|----------------------|---|
| (i) | $y = (f(x))^2$ | 1 |
| (ii) | $y = \sqrt{f(x)}$ | 1 |
| (iii) | $y = \frac{1}{f(x)}$ | 1 |

Question 15 continues on page 14

- (b) A particle of mass m is projected vertically upwards under gravity, the air resistance to the motion being $\frac{mgv^2}{k^2}$, where the speed is v , and k is a constant

- (i) Show that during the upward motion of the ball 2

$$\ddot{x} = -\frac{g}{k^2}(k^2 + v^2)$$

where x is the upward displacement.

- (ii) Hence, show that the greatest height reached is 3

$$\frac{k^2}{2g} \ln \left(1 + \frac{u^2}{k^2} \right)$$

where u is the speed of projection

- (c) An object is to undergo vertical motion on a bungee cord in a vacuum so that air resistance can be neglected. The only forces it experiences are gravity, mg and an elastic force, $-kmx$, where x is the particle's displacement from the origin. Initially, the object is at its lowest point given by $x = -a$. All constants are positive.

- (i) Using a force diagram show that 1

$$\ddot{x} = -g - kx$$

- (ii) By integration show that 3

$$v^2 = k \left(\left(a - \frac{g}{k} \right)^2 - \left(x + \frac{g}{k} \right)^2 \right)$$

- (iii) Show that the motion is described by 3

$$x = \left(\frac{g}{k} - a \right) \cos(\sqrt{k}t) - \frac{g}{k}$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) Let **2**

$$I_n = \int_0^1 x^n e^{-x} dx$$

Prove that for $n \geq 1$

$$I_n = nI_{n-1} - \frac{1}{e}$$

- (b) Use the binomial theorem to find the term independent of x in the expansion **3**

$$\left((x+1) + x^{-1}\right)^4$$

- (c) Find the cartesian equation of the locus of w where $w = \frac{z}{z+2}$ is purely imaginary. **2**

Question 16 continues on page 16

- (d) (i) Use de Moivre's theorem to show that 2

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$
- (ii) Deduce that $8x^3 - 6x - 1 = 0$ has solutions $x = \cos \theta$, 2
where $\cos 3\theta = \frac{1}{2}$
- (iii) Find the roots of $8x^3 - 6x - 1 = 0$ in the form $\cos \theta$ 2
- (iv) Hence, evaluate $\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$ 2

End of paper

2014 Ext 2 Trial

$$1) \int x \sec^2(x^2) dx = \frac{1}{2} \int 2x \sec^2(x^2) dx \\ = \frac{1}{2} \tan x^2 + C$$

B

2) $f'(x)$ has a double root @ $x = -2$ & so f has a triple root
Since f is increasing to right of stat pt @ $x = 4$
 f must be 

$$3) \sqrt{-3+4i} = \sqrt{1+4i-4} \\ = \sqrt{1+4i+4i^2} \\ = \sqrt{(1+2i)^2} \\ = 1+2i$$

B

4) Must be ~~at~~ hyperbola.
 $S = (ae, 0) \therefore e = \frac{3}{2}$
 $b^2 = 2^2 \left(\left(\frac{3}{2} \right)^2 - 1 \right)$
 $= 5$
 $\frac{x^2}{4} - \frac{y^2}{5} = 1$

C

5) $0 < y < \frac{\pi}{2}$

D

6) $e^x + e^y \frac{dy}{dx} = 0$
 $\frac{dy}{dx} = -\frac{e^x}{e^y}$
 $= -e^{x-y}$

A

7) $P(2) = 0 = 16 + 4a + 2b + 28$
 $P'(x) = 4x^2 + 2ax + b$
 $P'(2) = 32 + 4a + b = 0$
 $b = -12$
 $a = -5$

B

8) $\omega^6 + 1 = 0$
 $\therefore (\omega + 1)(\omega^5 - \omega^4 + \omega^3 - \omega^2 + \omega - 1)$ doesn't work for even power
 $\phi^5 - 1 = 0$
 $\therefore (\phi - 1)(\phi^4 + \phi^3 + \phi^2 + \phi + 1) = 0$
 $\therefore$ Only π C

9) $g(x) = x \tan x$ passes through origin D

10) D is even and $f(x)$ is always true
 $\therefore$ D

Question 11

$$\begin{aligned} \text{a) i) } \bar{z} - w &= 1 - i - (2 - 3i) \\ &= 1 - i - 2 + 3i \\ &= -1 + 2i \end{aligned}$$

$$\begin{aligned} \text{ii) } zw &= (1+i)(2-3i) \\ &= 2 - 3i + 2i - 3i^2 \\ &= 5 - i \end{aligned}$$

$$\begin{aligned} \text{iii) } |z| &= \sqrt{2} \quad \arg(z) = \frac{\pi}{4} \\ z &= \sqrt{2} \operatorname{cis} \left(\frac{\pi}{4} \right) \end{aligned}$$

$$\begin{aligned} \text{b) } \text{LHS} &= \left(\sqrt{2} \operatorname{cis} \left(\frac{\pi}{3} \right) \right)^8 - 8 \left(\sqrt{2} \operatorname{cis} \left(\frac{\pi}{3} \right) \right)^4 \\ &= 16 \operatorname{cis} \frac{8\pi}{3} - 16 \operatorname{cis} \frac{2\pi}{3} \\ &= 16 \operatorname{cis} \frac{2\pi}{3} - 16 \operatorname{cis} \frac{2\pi}{3} \\ &= 0 \\ &= \text{RHS} \end{aligned}$$

c) Since one root is $\sqrt{2}i$ another is $-\sqrt{2}i$
let other two roots be α, β
 $\alpha + \beta + \sqrt{2}i + (-\sqrt{2}i) = 1$

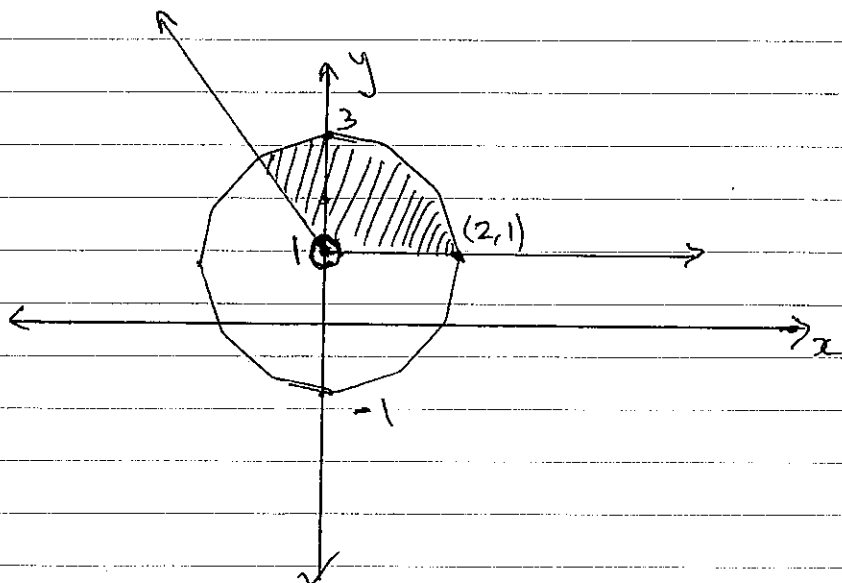
$$\alpha + \beta = 1$$

$$\alpha\beta + \sqrt{2}i + (-\sqrt{2}i) = -12$$

$$\alpha\beta = -6$$

$\therefore \alpha = 3$ and $\beta = -2$ are the other roots
 $\therefore$ real roots are 3 and -2

d)



$$e) i) \begin{aligned} \overrightarrow{PQ} &= z_2 - z_1 \\ \overrightarrow{QR} &= z_2 + i(z_2 - z_1) - z_2 \\ &= i(z_2 - z_1) \end{aligned}$$

Since multiplication by i rotates through 90° & doesn't change modulus

PQ is perpendicular to QR & equal in length

$\therefore PQR$ is right & isosceles

ii)

~~$$\overrightarrow{OS} = \overrightarrow{OP} + \overrightarrow{OQ}$$~~

$$\overrightarrow{OS} = \overrightarrow{OP} + \overrightarrow{QR}$$

$$= z_1 + z_2 - z_2 + i(z_2 - z_1)$$

$$= z_1 + i(z_2 - z_1)$$

Question 12

$$a) \int \frac{dx}{\sqrt{6 - (x^2 + 4x + 4)}} = \int \frac{dx}{\sqrt{6 - (x+2)^2}}$$

$$= \sin^{-1}\left(\frac{x+2}{\sqrt{6}}\right) + C$$

$$b) \int \frac{\sin^2 x}{\cos^2 x} \times \sin x dx = \int \frac{(1 - \cos^2 x) \sin x}{\cos^2 x} dx$$

$$= \int \frac{\sin x}{\cos^2 x} - \frac{\cos^2 x \sin x}{\cos^2 x} dx$$

$$= \int \sec x \tan x - \sin x dx$$

by Std Integrals $= \sec x + \cos x + C$

$$c) \int_0^1 \frac{2dt}{(1+t^2) \left(\frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2+1} \right)}$$

$$= \frac{1}{2} \int_0^1 \frac{2dt}{1-t^2+2t+1+t^2}$$

$$= \frac{1}{2} \int_0^1 \frac{2dt}{2+2t}$$

$$= \left[\ln(2+2t) \right]_0^1$$

$$= \ln 4 - \ln 2$$

$$= \ln 2$$

$$t = \tan\left(\frac{x}{2}\right)$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$\frac{dt}{dx} = \frac{1}{2} (1 + \tan^2\left(\frac{x}{2}\right))$$

$$\frac{2dt}{1+t^2} = dx$$

when $x = \frac{\pi}{2}$ $t = 1$
 $x = 0$ $t = 0$

$$d) i) 5x^3 - 3x^2 + 2x - 1 = A x^2(x+1) + B x(x^2+1) + (Cx+D)x^2$$

let $x=0$ $-1 = B$

$$= Ax^3 + Ax^2 - x^2 - 1 + Cx^3 + Dx^2$$

$$= (A+C)x^3 + (D-1)x^2 + Ax - 1$$

by equating coeffs. $A = 2$

$$D-1 = -3 \Rightarrow D = -2$$

$$C = 3$$

$$\begin{aligned}
 \text{ii)} \int \frac{5x^3 - 3x^2 + 2x - 1}{x^4 + x^2} dx &= \int \frac{2}{x} - \frac{1}{x^2} + \frac{3x-2}{x^2+1} dx \\
 &= \int \frac{2}{x} - \frac{1}{x^2} + \frac{3x}{x^2+1} - \frac{2}{x^2+1} dx \\
 &= 2 \ln x + \cancel{2x} + \frac{3}{2} \ln(x^2+1) - 2 \tan^{-1} x + C
 \end{aligned}$$

$$\begin{aligned}
 \text{e)} \int_1^e \frac{\ln x}{\sqrt{x}} dx &= \cancel{2\sqrt{x} \ln x} \quad \begin{array}{l} u = \ln x \\ u' = \frac{1}{x} \end{array} \quad \begin{array}{l} v' = x^{-1/2} \\ v = 2x^{1/2} \end{array} \\
 &= 2\sqrt{x} \ln x \Big|_1^e - \int_1^e 2x^{-1/2} dx \\
 &= 2\sqrt{e} \ln e - 2 \ln 1 - 2 \left[2x^{1/2} \right]_1^e \\
 &= 2\sqrt{e} - 2(2\sqrt{e} - 2) \\
 &= 2\sqrt{e} - 4\sqrt{e} + 4 \\
 &= 4 - 2\sqrt{e}
 \end{aligned}$$

Question 13

a) $\int_0^t f(x) dx = \int_t^0 f(t-y) (-dy)$ $\begin{matrix} x = t-y \\ \frac{dx}{dy} = -1 \end{matrix}$
 $= -\int_t^0 f(t-y) dy$ when $\begin{matrix} x = t \\ x = 0 \end{matrix}$ $\begin{matrix} y = 0 \\ y = t \end{matrix}$
 $= \int_0^t f(t-y) dy$

Since y is just a dummy variable
 $= \int_0^t f(t-x) dx$

ii) $\int_0^1 x(1-x)^{2014} dx = \int_0^1 (1-x)x^{2014} dx$
 $= \int_0^1 x^{2014} - x^{2015} dx$
 $= \left[\frac{1}{2015} x^{2015} - \frac{1}{2016} x^{2016} \right]_0^1$
 $= \frac{1}{2015} \times 1 - \frac{1}{2016} \times 1 - (0)$
 $= \frac{1}{4062240}$

b) i) $x = a \cos \theta$ $y = b \sin \theta$
 $\frac{dx}{d\theta} = -a \sin \theta$ $\frac{dy}{d\theta} = b \cos \theta$
 $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$
 $= \frac{b \cos \theta}{-a \sin \theta}$

$y - b \sin \theta = \frac{b \cos \theta}{-a \sin \theta} (x - a \cos \theta)$
 $-ay \sin \theta + ab \sin^2 \theta = b \cos \theta x - ab \cos^2 \theta$
 $0 = (b \cos \theta)x + (a \sin \theta)y - ab(\sin^2 \theta + \cos^2 \theta)$
 $0 = (b \cos \theta)x + (a \sin \theta)y - ab$

$$\text{ii) } S = (ae, 0) \quad S' = (-ae, 0)$$

$$QS = \left| \frac{abe \cos \theta - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right|$$

$$RS' = \left| \frac{-abe \cos \theta - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right|$$

$$\begin{aligned} SQ \times S'R &= \left| \frac{ab(e \cos \theta - 1)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \times \frac{-ab(e \cos \theta + 1)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right| \\ &= \left| \frac{-a^2 b^2 (e^2 \cos^2 \theta - 1)}{b^2 \cos^2 \theta + a^2 \sin^2 \theta} \right| \end{aligned}$$

for an ellipse
 $b^2 = a^2(1 - e^2)$

$$\begin{aligned} &= \left| \frac{-a^2 b^2 (e^2 \cos^2 \theta - 1)}{a^2 (1 - e^2) \cos^2 \theta + a^2 \sin^2 \theta} \right| \\ &= \left| \frac{-a^2 b^2 (e^2 \cos^2 \theta - 1)}{a^2 \cos^2 \theta - a^2 e^2 \cos^2 \theta + a^2 \sin^2 \theta} \right| \\ &= \left| \frac{-a^2 b^2 (e^2 \cos^2 \theta - 1)}{a^2 (\cos^2 \theta + \sin^2 \theta) - a^2 e^2 \cos^2 \theta} \right| \\ &= \left| \frac{a^2 b^2 (1 - e^2 \cos^2 \theta)}{a^2 (1 - e^2 \cos^2 \theta)} \right| \\ &= b^2 \end{aligned}$$

$$\text{c) i) } T_1 = (ct_1, \frac{c}{t_1})$$

$$m_{T_1} = \frac{\frac{c}{t_1} - \frac{c}{t_2}}{ct - ct_1}$$

$$= \frac{ct_1 - ct}{t_1 t}$$

$$\begin{aligned} &= \frac{c(t - t_1)}{t_1 t} \div \frac{c(t - t_1)}{1} \\ &= -\frac{1}{t_1 t} \end{aligned}$$

Slope for $TT_2 = -\frac{1}{t_2 t}$
 Since perpendicular

$$-\frac{1}{t_2 t} \times -\frac{1}{t_1 t} = -1$$

$$\frac{1}{t_1 t_2 t^2} = -1$$

$$t^2 = -\frac{1}{t_1 t_2}$$

$$ii) M_{T_1 T_2} = -\frac{1}{t_1 t_2}$$

$$iii) y = \frac{c^2}{x}$$
$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$@ T \text{ slope is } -\frac{c^2}{(ct)^2} = -\frac{1}{t^2}$$

$\therefore$ Normal @ T has slope t^2

$$\text{From i) } t^2 = -\frac{1}{t_1 t_2}$$

$\therefore$ normal parallel to $T_2 T_1$

Question 14

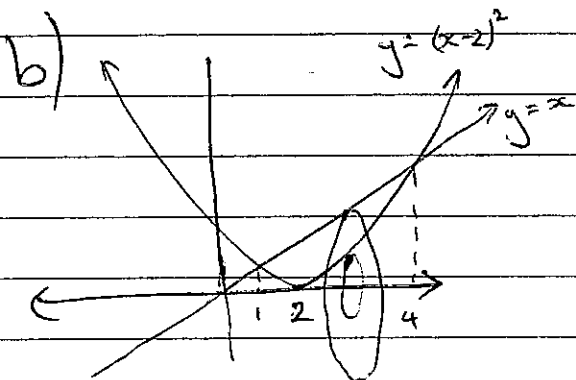
a) let $x = z - 2$

$$z = x + 2$$

$$(x+2)^3 - 6(x+2)^2 + 12(x+2) - 35 = 0$$

$$x^3 + 6x^2 + 12x + 8 - 6x^2 - 24x - 24 + 12x + 24 - 35 = 0$$

$$x^3 - 27 = 0$$



$$(x-2)^2 = x$$

$$x^2 - 4x + 4 = x$$

$$x^2 - 5x + 4 = 0$$

$$(x-4)(x-1) = 0$$

$$x = 1 \text{ or } 4$$

an element of volume is

$$\Delta V = \pi (y_1^2 - y_2^2) \Delta x$$

$$= \pi (x^2 - (x-2)^2) \Delta x$$

$$y_1 = x \quad y_2 = (x-2)^2$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=1}^4 \pi (x^2 - (x-2)^2) \Delta x$$

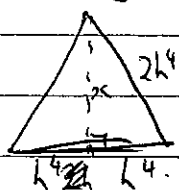
$$= \pi \int_1^4 (x^2 - (x-2)^2) dx$$

$$= \pi \left[\frac{2x^3}{3} - 4x^2 \right]_1^4$$

$$= \pi (32 - 16 - (2 - 4))$$

$$=$$

c) i) @ $x = h$
length of all sides $2h$



$$(2h)^2 = x^2 + (h)^2$$

$$4h^2 = x^2 + h^2$$

$$x^2 = 3h^2$$

$$x = \sqrt{3} h$$

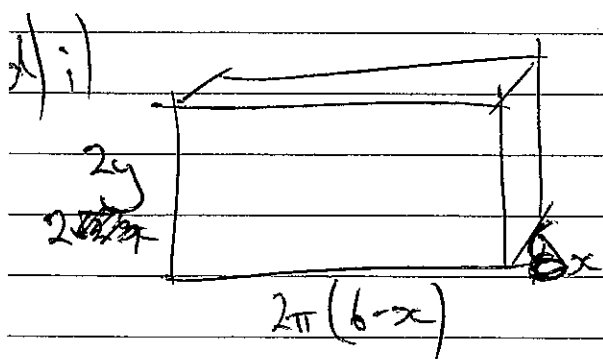
$$\text{Area} = \frac{1}{2} \times 2h \times \sqrt{3} h$$

$$= \sqrt{3} h^2$$

ii) $V = \int_0^2 \sqrt{3} x^2 dx$

$$= \sqrt{3} \left[\frac{x^3}{3} \right]_0^2$$

$$= \frac{512\sqrt{3}}{9}$$



An element of volume is

$$\delta V = 2\sqrt{x} \times 2\pi(6-x) \times \delta x$$

$$= 8\pi \sqrt{x} (6-x) \delta x$$

ii) $V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^2 8\pi \sqrt{x} (6-x) \delta x$

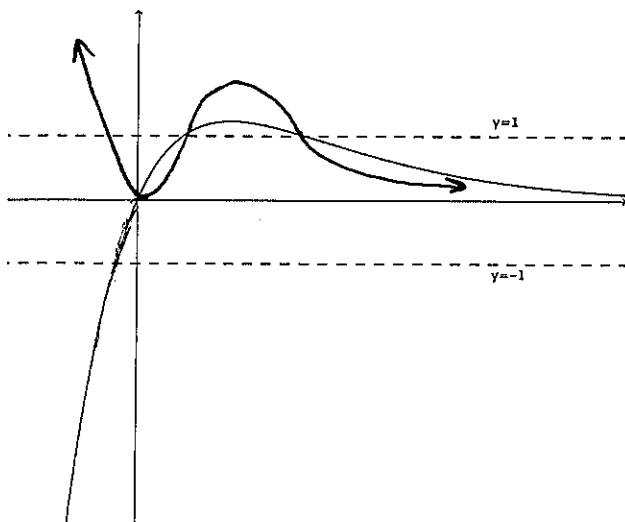
$$= 8\pi \int_0^2 \sqrt{x} (6-x) dx$$

$$\begin{aligned}
 \text{ii)} \quad V &= 8\pi \int_0^2 6x^{1/2} - x^{3/2} dx \\
 &= 8\pi \left[4x^{3/2} - \frac{2}{5}x^{5/2} \right]_0^2 \\
 &= 8\pi \left(8\sqrt{2} - \frac{8\sqrt{2}}{5} - (0) \right) \\
 &= \frac{256\sqrt{2}\pi}{5}
 \end{aligned}$$

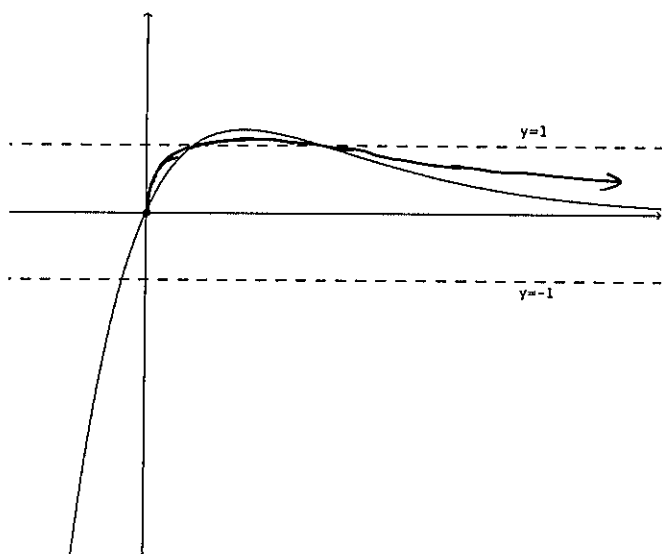
Question 15 a Answer Sheet

Insert into question 15 answer booklet

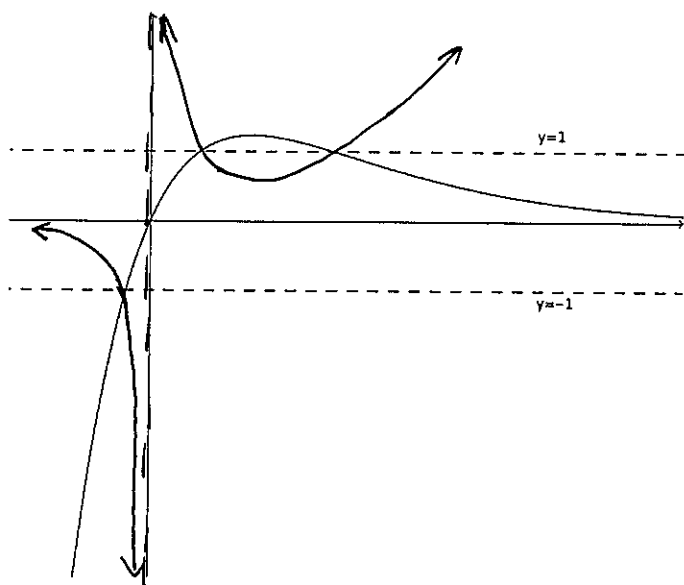
(i) $y = (f(x))^2$



(ii) $y = \sqrt{f(x)}$



(iii) $y = \frac{1}{f(x)}$



Qn 15

b) i)

$$F = -mg - \frac{mgv^2}{k^2}$$

$$\frac{-mgv^2}{k^2} \downarrow \ominus \downarrow -mg \quad m \ddot{x} = -mg \left(\frac{k^2 + v^2}{k^2} \right)$$
$$= -\frac{mg}{k^2} (k^2 + v^2)$$

ii) $v \frac{dv}{dx} = -\frac{mg}{k^2} (k^2 + v^2)$

$$\frac{1}{2} \frac{2v}{k^2 + v^2} dv = -\frac{mg}{k^2} dx$$

$$\frac{1}{2} \ln(k^2 + v^2) = -\frac{mg}{k^2} x + C$$

@ $x=0 \quad v=u$

$$\frac{1}{2} \ln(k^2 + u^2) = C$$

max height when $v=0$

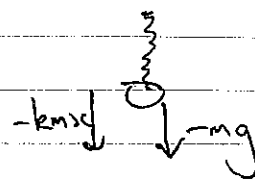
$$\frac{1}{2} \ln(k^2) = -\frac{g}{k^2} x + \frac{1}{2} \ln(k^2 + u^2)$$

$$\frac{g}{k^2} x = \frac{1}{2} \left(\ln(k^2 + u^2) - \ln k^2 \right)$$

$$x = \frac{k^2}{2g} \ln \left(\frac{k^2 + u^2}{k^2} \right)$$

$$= \frac{k^2}{2g} \ln \left(1 + \frac{u^2}{k^2} \right)$$

$$\begin{aligned}
 \text{c) i) } F &= -mg - kx \\
 m\ddot{x} &= -mg - kx \\
 m\ddot{x} &= m(-g - kx) \\
 \ddot{x} &= -g - kx
 \end{aligned}$$



$$\begin{aligned}
 \text{ii) } v \frac{dv}{dx} &= -g - kx \\
 \frac{1}{2} v^2 &= -gx - \frac{k}{2} x^2 + C
 \end{aligned}$$

$$\begin{aligned}
 @ \ x = -a \quad v &= 0 \quad (\text{initially stationary at limit of motion}) \\
 0 &= ag - \frac{a^2 k}{2} + C \\
 C &= \frac{a^2 k}{2} - ag
 \end{aligned}$$

$$\frac{1}{2} v^2 = -gx - \frac{k}{2} x^2 + \frac{a^2 k}{2} - ag$$

$$\begin{aligned}
 v^2 &= a^2 k - 2ag - 2gx - kx^2 \\
 &= k \left(a^2 - 2a \frac{g}{k} - \frac{2gx}{k} - x^2 \right) \\
 &= k \left(a^2 - \frac{2ag}{k} + \frac{g^2}{k^2} - \frac{g^2}{k^2} - \frac{2gx}{k} - x^2 \right) \\
 &= k \left(\left(a - \frac{g}{k} \right)^2 - \left(\frac{g^2}{k^2} + \frac{2gx}{k} + x^2 \right) \right) \\
 &= k \left(\left(a - \frac{g}{k} \right)^2 - \left(x + \frac{g}{k} \right)^2 \right)
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } v &= \pm \sqrt{k} \sqrt{\left(a - \frac{g}{k} \right)^2 - \left(x + \frac{g}{k} \right)^2} \\
 \frac{-dx}{\sqrt{\left(a - \frac{g}{k} \right)^2 - \left(x + \frac{g}{k} \right)^2}} &= \pm \sqrt{k} dt
 \end{aligned}$$

$$\cos^{-1} \left(\frac{x + \frac{g}{k}}{a - \frac{g}{k}} \right) = \pm \sqrt{k} t + C$$

$$@ \ t = 0 \quad x = -a$$

$$\cos^{-1} \left(\frac{-a + \frac{g}{k}}{a - \frac{g}{k}} \right) = \pm \sqrt{k} \times 0 + C$$

$$\cos^{-1}(-1) = C$$

$$C = \pi$$

$$\frac{x + \frac{g}{k}}{a - \frac{g}{k}} = \cos \left(\pi \pm \sqrt{k} t \right)$$

Since $\cos(\pi + \alpha) = -\cos(\pi - \alpha)$ we can take either without any concern.
choosing the $-\cos$

$$\frac{x + \frac{g}{R}}{a - \frac{g}{R}} = \cos(\pi - \sqrt{k}t)$$

Since $\cos(\pi - \alpha) = -\cos \alpha$ for all α

$$\frac{x + \frac{g}{R}}{a - \frac{g}{R}} = -\cos \sqrt{k}t$$

$$x + \frac{g}{R} = \left(\frac{g}{R} - a \right) \cos \sqrt{k}t$$

$$x = \left(\frac{g}{R} - a \right) \cos \sqrt{k}t - \frac{g}{R}$$

Question 16

$$a) I_n = \int_0^1 x^n e^{-x} dx$$

$$u = x^n \quad u' = nx^{n-1}$$

$$v' = e^{-x} \quad v = -e^{-x}$$

$$= \left[x^n e^{-x} \right]_0^1 - \int_0^1 nx^{n-1} e^{-x} dx$$

$$= 1^n e^{-1} - (0^n e^{-0}) + n \int_0^1 x^{n-1} e^{-x} dx$$

$$= n \int_0^1 x^{n-1} e^{-x} dx - \frac{1}{e}$$

$$= n I_{n-1} - \frac{1}{e}$$

$$b) \left((x+1) + x^{-1} \right)^4 = \sum_{r=0}^4 {}^4C_r (x+1)^{4-r} (x^{-1})^r$$

Using binomial theorem on $(x+1)^{4-r} = \sum_{k=0}^{4-r} {}^{4-r}C_k x^k x^{4-r-k}$

$$\left((x+1) + x^{-1} \right)^4 = \sum_{r=0}^4 {}^4C_r x^{-r} \times \sum_{k=0}^{4-r} {}^{4-r}C_k x^k$$

for term independent of x

$$x^{-r} \times x^k = x^0$$

$$-r + k = 0$$

$$r = k$$

∴ const term is

$$\sum_{r=0}^4 {}^4C_r \times \sum_{r=0}^{4-r} {}^{4-r}C_r = {}^4C_0 \times {}^4C_0 + {}^4C_1 \times {}^3C_1 + {}^4C_2 \times {}^2C_2 = 19$$

c) let $w = x + iy$
to be purely imaginary real part = 0
 $x = 0$

but 0 is not an imaginary number so
locus is

$$x = 0 \text{ except } (0,0)$$

d) i) consider

$$\begin{aligned}\cos 3\theta &= (\cos \theta)^3 \\ &= (\cos \theta + i \sin \theta)^3 \\ &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta + 3i^2 \cos \theta \sin^2 \theta + i^3 \sin^3 \theta \\ &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta\end{aligned}$$

equating real parts

$$\begin{aligned}\cos 3\theta &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\ &= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta\end{aligned}$$

ii) when $\cos 3\theta = \frac{1}{2}$

$$4 \cos^3 \theta - 3 \cos \theta = \frac{1}{2}$$

$$8 \cos^3 \theta - 6 \cos \theta = 1$$

$$8 \cos^3 \theta - 6 \cos \theta - 1 = 0$$

which is $8x^3 - 6x - 1 = 0$ when $x = \cos \theta$

$\therefore$ the solutions are given by $x = \cos \theta$

iii) the roots of $8x^3 - 6x - 1 = 0$

are given by the solutions of $\cos 3\theta = \frac{1}{2}$

$$3\theta = 2\pi n \pm \frac{\pi}{3}$$

$$\theta = \frac{2n\pi}{3} \pm \frac{\pi}{9}$$

for $n=0$ $\theta = \frac{\pi}{9}$ or $\frac{5\pi}{9}$ taking +ve

$$n=1 \quad \theta = \frac{7\pi}{9}$$

$$n=2 \quad \theta = \frac{13\pi}{9}$$

only need 3 roots since degree 3 polynomial

$\therefore$ roots are

$$\cos \frac{\pi}{9}, \cos \frac{13\pi}{9}, \cos \frac{7\pi}{9}$$

$$iv) \cos \frac{7\pi}{9} = \cos \frac{2\pi}{9}$$

$$= \cos \frac{2\pi}{9}$$

$$\therefore \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$$

$$= \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$$

subtracting 2π

$\cos$ is even function

$\therefore$ the product of the roots.

iv) roots are $\cos \frac{\pi}{9}$, $\cos \frac{7\pi}{9}$ and $\cos \frac{13\pi}{9}$

consider $\cos \frac{7\pi}{9}$.

$$\begin{aligned}\cos \frac{7\pi}{9} &= \cos \left(\pi - \frac{2\pi}{9} \right) \\ &= -\cos \frac{2\pi}{9}\end{aligned}$$

$$\begin{aligned}\text{consider } \cos \frac{13\pi}{9} &= \cos \left(-\frac{5\pi}{9} \right) \\ &= \cos \left(-\pi + \frac{4\pi}{9} \right) \\ &= -\cos \frac{4\pi}{9}\end{aligned}$$

$\therefore$ roots are $\cos \frac{\pi}{9}$, $-\cos \frac{2\pi}{9}$ and $-\cos \frac{4\pi}{9}$

$\therefore$ product of roots is

$$\begin{aligned}\cos \frac{\pi}{9} \times -\cos \frac{2\pi}{9} \times -\cos \frac{4\pi}{9} \\ = \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}\end{aligned}$$

product of roots is $\frac{1}{8}$

$$\therefore \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$$